

How Glacial Processes Control Southern Ocean Productivity Across Glacial Cycles

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Introduction

Glaciers supply bioavailable iron to the Southern Ocean, where phytoplankton depend on iron for growth and photosynthesis. For a long time, the fertilization potential of glaciers was measured by total iron flux alone, but this overlooked iron speciation (Shoenfelt et al., 2018b). Speciation matters because the glacial iron supply has been found to correspond to a 50–60 ppm decrease in atmospheric CO₂ at the onset of glaciation (Struve et al., 2026). As glaciers retreat under a warming climate, this process is likely to change.

1 Iron Speciation

Iron is essential to phytoplankton photosynthesis and nitrogen assimilation. Photosystem I contains up to 12 iron atoms, and the enzymes that convert nitrate into usable nitrogen are also iron-dependent (Schoffman et al., 2016). Without iron, phytoplankton cannot photosynthesize or grow.

Not all iron is equally usable. Soluble Fe(II) is the reduced form that dominated early, low-oxygen Earth and remains stable in modern reducing environments. It is reactive enough for cells to absorb directly. Fe(III), by contrast, precipitates under oxic conditions as ferric oxyhydroxides that phytoplankton cannot access (Schoffman et al., 2016; Stimpfle et al., 2026). Stimpfle et al. (2026) showed experimentally that glacier meltwater, enriched in bioavailable iron, doubled phytoplankton iron uptake in iron-limited Southern Ocean conditions, while groundwater containing less soluble iron produced no fertilization response. Iron speciation, not quantity, determines phytoplankton activity.

2 Glacial Processes and Iron Mineralogy

Bedrock in glaciated regions typically consists of continental crustal assemblages, predominantly granite, basalt, gabbro, and gneiss (Wedepohl, 1995; Rudnick & Gao, 2014). Iron is incorporated into silicate lattices in these rocks, residing in olivine and pyroxene within mafic rocks, and in biotite or amphibole within granitic and metamorphic suites (Raiswell et al., 2012). Under reducing conditions, iron occurs as Fe(II) through substitution with magnesium, while Fe(III) partitions into distinct oxide phases only when higher oxygen levels are available during rock formation.

Glaciers exert basal stresses from the weight of the ice. The Greenland and Antarctic ice sheets reach thicknesses of 3–4 km, generating pressures above 30 MPa and driving abrasion and quarrying at the glacier bed (Cuffey & Paterson, 2010). As glaciers move downslope, rock fragments entrained in basal ice scrape and fracture the underlying substrate, producing glacial till. The finest fraction of this material, glacial flour (silt-sized particles smaller than 63 μm), exposes fresh iron-bearing mineral surfaces through mechanical comminution (Hallet et al., 1996).

Fe(II) persists beneath glaciers because meltwater is isolated from the atmosphere and

oxygen is rapidly consumed by chemical reactions and microbial respiration, creating reducing conditions in which dissolved ferrous iron accumulates (Wadham et al., 2010; Raiswell & Canfield, 2012; Hawkings et al., 2014). Microorganisms further enhance Fe(II) production by using ferric iron as an electron acceptor during respiration, increasing dissolved Fe(II) concentrations in subglacial waters (Nixon et al., 2017; Wadham et al., 2010).

3 Iron Path from Glaciers to the Ocean

At marine-terminating glaciers, subglacial meltwater discharges directly into the ocean. Its lower density drives buoyant freshwater plumes upward through the water column, entraining surrounding seawater and transporting iron-rich particles into the photic zone while drawing deeper, iron-enriched waters toward the surface. Remineralization of sinking organic matter further elevates iron concentrations in these upwelling zones (Wadham et al., 2019).

In land-terminating settings, meltwater forms proglacial rivers that carry suspended glacial flour across outwash plains toward fjords and coastal oceans. In these systems, iron arrives primarily in particulate and colloidal forms capable of retaining reduced Fe(II) during transport, rather than as dissolved Fe(II) (Hallet et al., 1996; Hawkings et al., 2014; Wadham et al., 2019). Fine glacial sediments exposed on outwash plains may also be mobilized as mineral dust and deposited into nearby marine environments via aeolian transport (Boyd et al., 2010).

4 Glacial–Interglacial Iron Supply

These processes operated at a much larger scale during glacial periods than they do today. The Patagonian Ice Sheet, at its last maximum, extended 2,090 km along the Andes Mountains, with outlet lobes reaching hundreds of kilometers onto the Argentine lowlands and the Pacific continental shelf edge. As the ice moved, it ground rock into fine sediments, deposited debris as moraines at its margins, and distributed glacially derived material across broad outwash plains. Today, the region holds only small, fragmented ice fields covering 22,700 km², less than 5% of the maximum ice-sheet coverage (Davies et al., 2020). The moraines and outwash plains that remain are a record of the Fe(II)-rich sediment production that occurred on a massive scale during glacial periods.

Shoenfelt et al. (2018b) quantified this signal using iron K-edge X-ray absorption spectroscopy on two sediment cores from the sub-antarctic region. This method estimates the oxidation state of iron by comparing the absorption spectrum of each sample to known Fe(II)/Fe(III) mineral ratios. The Fe(II)/Fe_{total} ratio changed from 5–10% during interglacial periods to 25–45% during glacial periods, a sixfold increase. To express the magnitude of change relative to a modern baseline, the authors used an amplification factor, defined as the ratio of iron flux at a given time to the Holocene baseline flux. Fe_{total} flux was amplified 3–5 times during glacial periods, but the Fe(II) fraction increased with an amplification factor of 15 to 20. Figure 1 shows this excess bioavailable iron supply over time, and Table 1 summarizes the statistics by period.

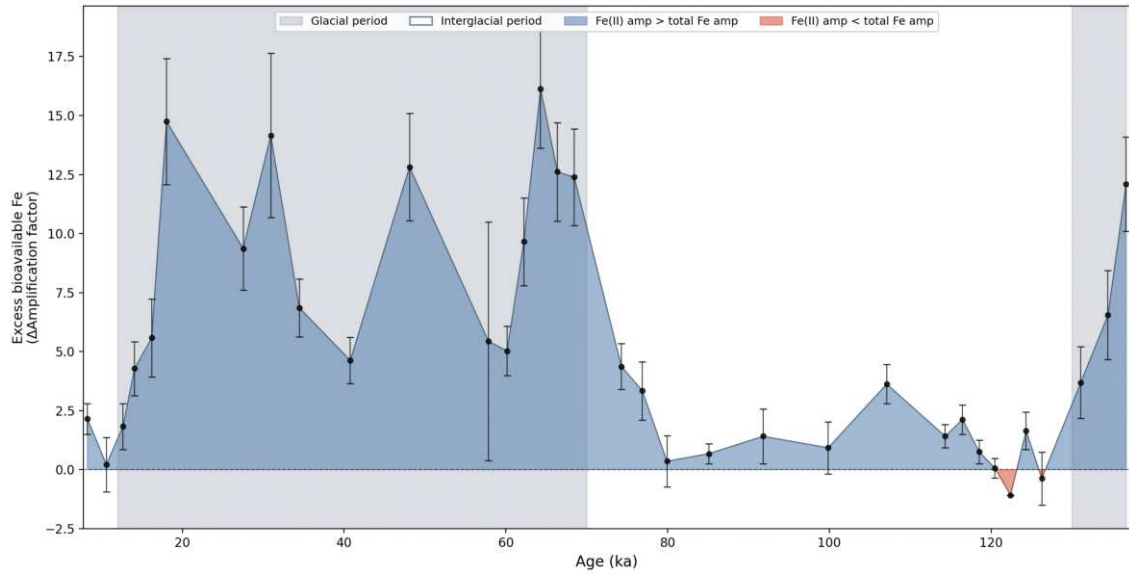


Figure 1: Excess bioavailable iron supply relative to total iron supply across two glacial cycles, PS75/56-1 sediment core, Southern Ocean. The y-axis shows the difference between the Fe(II) amplification factor and the Fe_{total} amplification factor, both calculated relative to the Holocene baseline flux. A value of zero means the two are equally elevated compared to the present. Positive values indicate excess bioavailable iron. Fe(II) amplification factors were determined using iron K-edge X-ray absorption spectroscopy on the PS75/56-1 core from the South Pacific Southern Ocean (Shoenfelt et al., 2018a; Shoenfelt et al., 2018b). Fe_{total} amplification factors were derived from thorium-230-normalized iron flux measurements on the same core. Error bars show propagated XAS measurement uncertainty. Grey shading marks two glacial periods: Marine Isotope Stage 2 (MIS 2, Last Glacial Maximum, ~12–70 ka) and Marine Isotope Stage 6 (MIS 6, Penultimate Glaciation, ~130–137 ka). The signal is consistently positive and larger during glacial periods, indicating that glacial weathering disproportionately increases bioavailable Fe(II) relative to Fe_{total} . Two data points near 122 and 126 ka show small negative values, possibly reflecting measurement uncertainty or the dominance of chemical over physical weathering during the preceding interglacial.

Table 1: Summary statistics of excess bioavailable iron supply by climate period. Δ = Fe(II) amplification factor minus Fe_{total} amplification factor (dimensionless), derived from XAS measurements on the PS75/56-1 sediment core, Southern Ocean (Shoenfelt et al., 2018a; Shoenfelt et al., 2018b). Both amplification factors are calculated on a Holocene baseline flux basis. SD = standard deviation; SE = standard error of the mean; Mean XAS uncertainty = average propagated measurement error per data point. Both glacial periods show mean excess values five to six times higher than the interglacial mean of 1.35. The high SD within MIS 2 (4.53) reflects variability in iron delivery across the glacial cycle rather than measurement noise, as it exceeds the mean XAS uncertainty of 2.05. MIS 6 has only three data points, so its statistics should be interpreted with caution.

Period	Age (ka)	n	Mean Δ	SD	SE	Mean XAS uncertainty
Interglacial (Holocene + MIS 5)	8.3–126.3	16	1.35	1.49	0.37	0.79
MIS 2 (Last Glacial Maximum)	12.7–68.5	15	9.04	4.53	1.17	2.05
MIS 6 (Penultimate Glaciation)	131.1–136.7	3	7.44	4.28	2.47	1.81

5 Future Implications

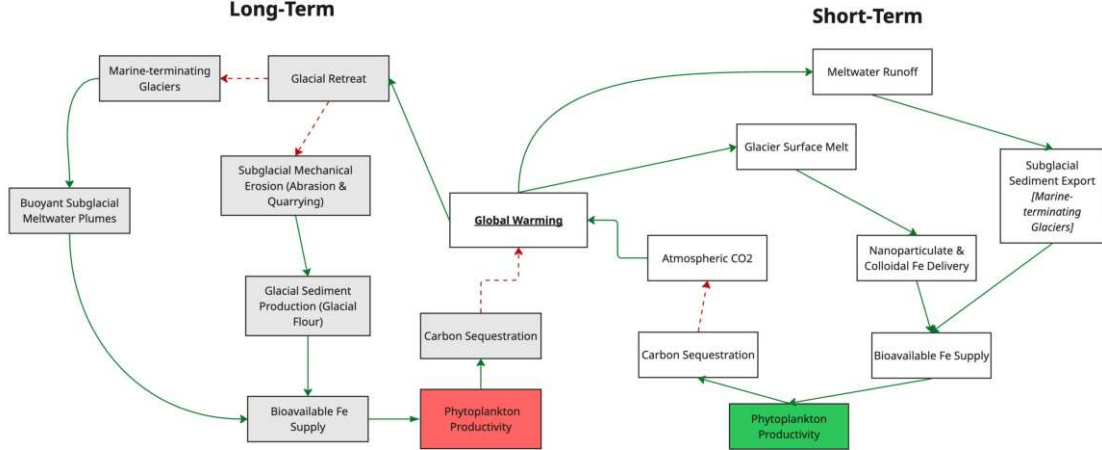


Figure 2: Conceptual model of short- and long-term glacier-iron-climate feedbacks. Green arrows indicate positive relationships; red dashed arrows indicate negative ones. In the short term, warming increases meltwater runoff and sediment export, enhancing delivery of bioavailable iron and phytoplankton productivity. In the long term, glacier retreat reduces marine-terminating glaciers and meltwater plumes, decreasing iron supply and potentially limiting productivity.

Anthropogenic climate change has increased global mean surface temperature by approximately 1.1°C above pre-industrial levels, primarily through greenhouse gas emissions (IPCC, 2023). Warming reduces glacier mass balance by increasing surface melt and meltwater

runoff, driving glacier retreat. Satellite observations show that global glaciers lost 6,542 Gt of ice between 2000 and 2023, equivalent to $273 \pm 16 \text{ Gt yr}^{-1}$ (Hugonnet et al., 2021). These changes alter glacier hydrology, increasing meltwater discharge, subglacial sediment export, and the delivery of nanoparticulate iron to marine environments via buoyant plumes at marine-terminating glaciers.

Figure 2 illustrates how glacier–iron–climate interactions play out across different timescales. In the short term, warming increases glacier melt and meltwater runoff, mobilizing subglacial sediments and delivering nanoparticulate and colloidal iron to surface waters. As noted earlier, meltwater can double phytoplankton iron uptake in iron-limited Southern Ocean conditions (Stimpfle et al., 2026). Increased phytoplankton productivity strengthens the biological carbon pump, raising carbon sequestration and lowering atmospheric CO_2 (Struve et al., 2026), forming a balancing feedback loop between ocean productivity and climate.

In the long term, warming causes glacier retreat, reducing marine-terminating glaciers and their iron-rich buoyant plumes. This decreases iron delivery and may limit phytoplankton productivity over longer timescales.

These dynamics depend on several standing conditions, including iron-bearing bedrock, subglacial microbial redox processes that maintain Fe(II), and iron limitation in Southern Ocean ecosystems. If glacier retreat passes a threshold at which marine-terminating glaciers disappear and land-terminating glaciers shrink substantially, the plume-driven iron delivery pathway may collapse and river-based iron transport may reduce as well, shifting the system from short-term iron fertilization to long-term iron limitation.

Conclusion

Glacial processes regulate the iron supply to the Southern Ocean in terms of both quantity and type. Physical weathering generates Fe(II)-enriched phases that phytoplankton can use, and the sediment record shows that this signal is amplified 15 to 20 times during glacial intervals. The influence of warming on this mechanism works in two directions: in the short term, melt-driven iron fertilization may enhance phytoplankton productivity and carbon sequestration, but substantial glacial retreat could eventually reduce Fe(II) supply and weaken that response.

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Appendix: Python Code and Output

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.patches import Patch

df =
    pd.read_excel('AmplificationFactorsExcelFileFormat/PS750561AmplificationFactors.xlsx',
        header=0)
df.columns = ['age_total', 'fe_amp', 'blank', 'age_feII', 'feII_amp',
    'feII_amp_error']
df = df.drop(columns=['blank'])

total = df[['age_total', 'fe_amp']].dropna()
feII = df[['age_feII', 'feII_amp', 'feII_amp_error']].dropna()

for col in ['age_total', 'fe_amp']:
    total[col] = pd.to_numeric(total[col], errors='coerce')
for col in ['age_feII', 'feII_amp', 'feII_amp_error']:
    feII[col] = pd.to_numeric(feII[col], errors='coerce')

total = total.dropna()
feII = feII.dropna()

total_sorted = total.sort_values('age_total')
total_interp = np.interp(feII['age_feII'], total_sorted['age_total'],
    total_sorted['fe_amp'])

feII['total_interp'] = total_interp
feII['difference'] = feII['feII_amp'] - feII['total_interp']
feII_sorted = feII.sort_values('age_feII').reset_index(drop=True)

feII_sorted['period'] = 'Interglacial'
feII_sorted.loc[(feII_sorted['age_feII'] >= 12) & (feII_sorted['age_feII'] <=
    70), 'period'] = 'MIS 2'
feII_sorted.loc[feII_sorted['age_feII'] >= 130, 'period'] = 'MIS 6'

# Print plotting data
print(f"{'Age (ka)':<12} {'Fe(II) amp':<14} {'Total Fe amp':<16}
    {'Difference':<14} {'Period'}")
print("-" * 70)
for _, row in feII_sorted.iterrows():
    print(f"{row['age_feII']:<12.3f} {row['feII_amp']:<14.4f}
        {row['total_interp']:<16.4f} {row['difference']:<14.4f} {row['period']}")

# Print stats
print("\nSUMMARY STATISTICS BY PERIOD")
period_full_names = {
    'Interglacial': 'Interglacial (Holocene + MIS 5)',
    'MIS 2': 'MIS 2 (Last Glacial Maximum)',
    'MIS 6': 'MIS 6 (Penultimate Glaciation)'
}
for period in ['Interglacial', 'MIS 2', 'MIS 6']:
    seg = feII_sorted[feII_sorted['period'] == period]
    n = len(seg)
```

```

mean_diff = seg['difference'].mean()
sd = seg['difference'].std(ddof=1)
se = sd / np.sqrt(n)
mean_xas_err = seg['feII_amp_error'].mean()
age_min = seg['age_feII'].min()
age_max = seg['age_feII'].max()
print(f"\n{period_full_names[period]}")
print(f" Age range: {age_min:.1f} to {age_max:.1f} ka")
print(f" n: {n}")
print(f" Mean Delta: {mean_diff:.3f}")
print(f" SD: {sd:.3f}")
print(f" SE: {se:.3f}")
print(f" Mean XAS error: {mean_xas_err:.3f}")

# Zero crossing
ages = feII_sorted['age_feII'].values
diffs = feII_sorted['difference'].values

new_ages, new_diffs = [], []
for i in range(len(ages)):
    new_ages.append(ages[i])
    new_diffs.append(diffs[i])
    if i < len(ages) - 1:
        if diffs[i] * diffs[i+1] < 0:
            zero_age = ages[i] + (0 - diffs[i]) * (ages[i+1] - ages[i]) /
                (diffs[i+1] - diffs[i])
            new_ages.append(zero_age)
            new_diffs.append(0.0)

new_ages = np.array(new_ages)
new_diffs = np.array(new_diffs)

x_min = feII_sorted['age_feII'].min()
x_max = feII_sorted['age_feII'].max()

# Figure
fig, ax = plt.subplots(figsize=(10, 4))

ax.axvspan(12, 70, alpha=0.25, color='slategray', label='Glacial period')
ax.axvspan(130, x_max, alpha=0.25, color='slategray')

interglacial_patch = Patch(facecolor='white', edgecolor='slategray',
    linewidth=1.2, label='Interglacial period')

ax.axhline(0, color='black', linewidth=0.8, linestyle='--', alpha=0.5)

ax.fill_between(new_ages, new_diffs, 0,
    where=(new_diffs >= 0),
    color='steelblue', alpha=0.55,
    label='Fe(II) amp > total Fe amp')
ax.fill_between(new_ages, new_diffs, 0,
    where=(new_diffs <= 0),
    color='tomato', alpha=0.6,
    label='Fe(II) amp < total Fe amp')

ax.errorbar(feII_sorted['age_feII'], feII_sorted['difference'],

```

```

yerr=feII_sorted['feII_amp_error'],
fmt='o', color='black', markersize=4,
capsize=3, linewidth=1.2, elinewidth=0.8,
ecolor='black', alpha=0.8, zorder=4)

ax.plot(feII_sorted['age_feII'], feII_sorted['difference'],
        '-', color='black', linewidth=0.8, alpha=0.4, zorder=3)

handles, labels = ax.get_legend_handles_labels()
legend_handles = [handles[0], interglacial_patch, handles[1], handles[2]]
ax.legend(handles=legend_handles, fontsize=9, loc='upper center',
          ncol=4, framealpha=0.9)

ax.set_xlabel('Age (ka)', fontsize=12)
ax.set_ylabel('Excess bioavailable Fe\n(Delta Amplification factor)',
              fontsize=11)
ax.set_xlim(x_min - 0.5, x_max + 0.5)

plt.tight_layout()
plt.savefig('shoenfelt_difference.png', dpi=300, bbox_inches='tight')
plt.show()

```

Code Output

Age (ka)	Fe(II) amp	Total Fe amp	Difference	Period

8.272	3.0281	0.8747	2.1534	Interglacial
10.612	1.4488	1.2320	0.2168	Interglacial
12.676	3.1840	1.3517	1.8324	MIS 2
14.097	6.0687	1.7851	4.2836	MIS 2
16.219	7.7544	2.1727	5.5817	MIS 2
18.052	17.7977	3.0524	14.7453	MIS 2
27.533	12.2135	2.8433	9.3702	MIS 2
30.942	16.6462	2.4909	14.1553	MIS 2
34.470	9.0940	2.2381	6.8559	MIS 2
40.774	6.3128	1.6867	4.6261	MIS 2
48.099	14.9779	2.1580	12.8198	MIS 2
57.865	7.3621	1.9259	5.4362	MIS 2
60.160	7.0178	1.9878	5.0301	MIS 2
62.233	11.4801	1.8270	9.6531	MIS 2
64.307	18.2621	2.1365	16.1255	MIS 2
66.381	14.6826	2.0675	12.6151	MIS 2
68.455	14.2295	1.8404	12.3891	MIS 2
74.300	5.7014	1.3289	4.3725	Interglacial
76.886	4.2869	0.9486	3.3383	Interglacial
79.925	1.2796	0.9225	0.3571	Interglacial
85.098	1.5470	0.8705	0.6765	Interglacial
91.815	2.4700	1.0574	1.4126	Interglacial
99.863	1.9184	0.9914	0.9270	Interglacial
107.092	4.6604	1.0379	3.6225	Interglacial
114.328	2.4071	0.9894	1.4177	Interglacial
116.469	2.9205	0.7935	2.1269	Interglacial
118.537	1.4171	0.6628	0.7543	Interglacial
120.470	0.8129	0.7492	0.0636	Interglacial
122.403	0.0000	1.0770	-1.0770	Interglacial

124.337	2.8942	1.2464	1.6478	Interglacial
126.270	1.4074	1.7793	-0.3719	Interglacial
131.098	5.8297	2.1399	3.6898	MIS 6
134.452	10.5152	3.9694	6.5458	MIS 6
136.692	14.6350	2.5388	12.0963	MIS 6

SUMMARY STATISTICS BY PERIOD

Interglacial (Holocene + MIS 5)

Age range: 8.3 to 126.3 ka
n: 16
Mean Delta: 1.352
SD: 1.491
SE: 0.373
Mean XAS error: 0.786

MIS 2 (Last Glacial Maximum)

Age range: 12.7 to 68.5 ka
n: 15
Mean Delta: 9.035
SD: 4.532
SE: 1.170
Mean XAS error: 2.050

MIS 6 (Penultimate Glaciation)

Age range: 131.1 to 136.7 ka
n: 3
Mean Delta: 7.444
SD: 4.275
SE: 2.468
Mean XAS error: 1.805

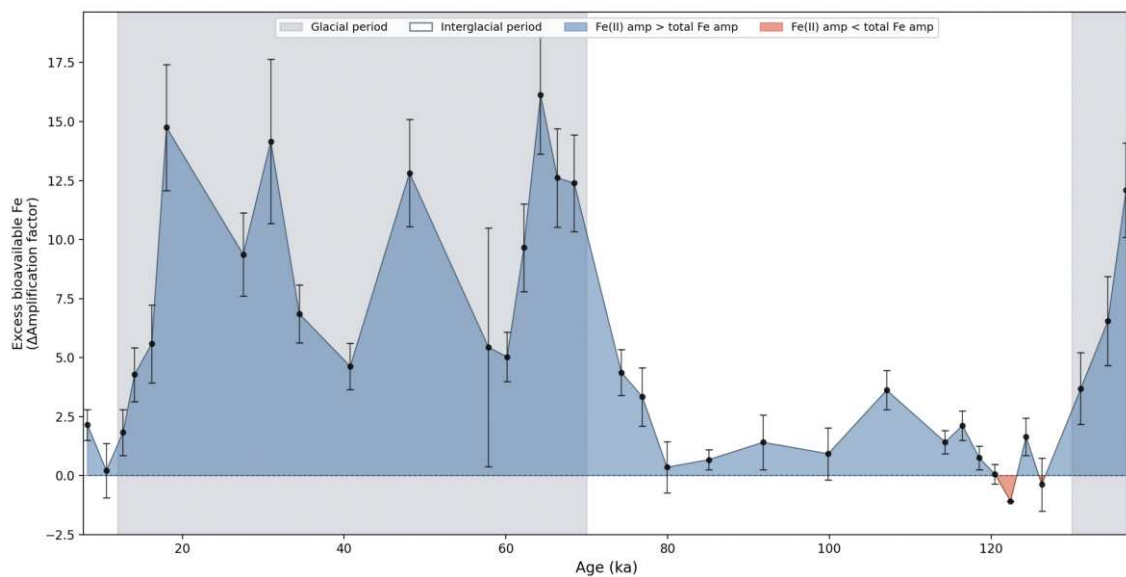


Figure output from Python script (see Figure 1 in main text).